Math 110 — Week 2

Linear independence, span, bases, dimension (Notes Ch. 3–4, LADR 2A–2C, HW2)

§1 Facts used from Week 1

(W1) A subset $U \subseteq V$ is a subspace if $0 \in U$ and $\lambda v + w \in U$ for all $v, w \in U$, $\lambda \in \mathbb{F}$. Intersections and sums $U + W = \{u + w\}$ of subspaces are subspaces.

(W2) The sum $U_1 + \cdots + U_m$ is direct if $U_k \cap (U_1 + \cdots + U_{k-1}) = \{0\}$ for each k. Equivalent: the only way to write $0 = u_1 + \cdots + u_m$ with $u_k \in U_k$ is with all $u_k = 0$ (Prop. 7). For two subspaces: $U \cap W = \{0\}$.

(W3) If $av = 0$ then $a = 0$ or $v = 0$.

§2 This week

§2.1 Linear independence, span, basis (Ch. 3)

The question. When does a finite list of vectors describe a whole vector space? Two things are needed: every vector can be built from the list, and in only one way. In $\mathbb{R}^2$: $(1,0), (0,1)$ does both. $(1,0), (0,1), (1,1)$ builds everything but not uniquely: $(1,1) = 1 \cdot (1,0) + 1 \cdot (0,1) = 1 \cdot (1,1)$. $(1,0)$ alone builds uniquely but misses $(0,1)$.

Definition (Def. 7, 8 / LADR 2.4, 2.15, 2.26, 2.29)**.** Let $v_1, \ldots, v_m \in V$. A **linear combination** is a sum $\lambda_1 v_1 + \cdots + \lambda_m v_m$ with $\lambda_k \in \mathbb{F}$.

- $\operatorname{span}(v_1, \ldots, v_m)$ is the set of all linear combinations.
- The list **spans** V if $\operatorname{span}(v_1, \ldots, v_m) = V$.
- The list is **linearly independent** if $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ forces every $\lambda_k = 0$. Otherwise it is **linearly dependent**.
- The list is a **basis** of V if it is linearly independent and spans V.
- V is **finite-dimensional** if some finite list spans it.

- The **length** of a list is its number of entries. The empty list has length 0, is linearly independent, and spans $\{0\}$.
- A list containing 0 is dependent: $1 \cdot 0 = 0$ with coefficient $1 \neq 0$.
- $\operatorname{span}(v_1, \ldots, v_m) = \mathbb{F}v_1 + \cdots + \mathbb{F}v_m$, where $\mathbb{F}v = \{\lambda v\}$. It is the smallest subspace containing the v_k (W1).
- Ex. 10: $e_1, \ldots, e_n$ is a basis of $\mathbb{F}^n$. The span of $(1, 1, 1, \ldots)$ and $(0, 1, 2, 3, \ldots)$ in $\mathbb{R}^\infty$ is the set of arithmetic sequences $(a_0, a_0 + d, a_0 + 2d, \ldots)$. The only basis of $\{0\}$ is the empty list.

Proposition (Basis = unique representation, LADR 2.28)

$v_1, \ldots, v_m$ is a basis of $V \iff$ every $v \in V$ can be written as $\lambda_1 v_1 + \cdots + \lambda_m v_m$ in exactly one way.

The idea

Spanning says a representation exists. Independence is what makes it unique, and the reason is one subtraction: two representations of the same vector differ by a representation of 0.

The notes package this with the map $\Phi\colon \mathbb{F}^m \to V$, $\Phi(\lambda_1, \dots, \lambda_m) = \lambda_1 v_1 + \cdots + \lambda_m v_m$: the list spans exactly when Φ is surjective, is independent exactly when Φ is injective, and is a basis exactly when Φ is bijective. The inverse of Φ sends a vector to its coordinates.

Proof. 1. ($\Rightarrow$) Existence is spanning. If $v = \sum \lambda_k v_k = \sum \mu_k v_k$, subtract: $0 = \sum(\lambda_k - \mu_k)v_k$. Independence gives $\lambda_k = \mu_k$ for all k.

2. ($\Leftarrow$) Existence of representations is spanning. $0 = 0v_1 + \cdots + 0v_m$ is one representation of 0; by uniqueness it is the only one, which is independence.

□

- Same statement in terms of sums: $v_1, \dots, v_m$ (all nonzero) is independent $\iff$ the sum $\mathbb{F}v_1 + \cdots + \mathbb{F}v_m$ is direct (W2 and W3).

Proposition (Prop. 8 / LADR 2.19: linear dependence lemma)

Let $v_1, \dots, v_m$ be linearly dependent. Then

(i) there is an index j with $v_j \in \operatorname{span}(v_1, \dots, v_{j-1})$;

(ii) for any such j, removing v_j does not change the span.

The idea

Dependence gives an equation $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ with some nonzero coefficient. Any vector with a nonzero coefficient can be solved for in terms of the others; that vector is redundant.

Why choose the *last* nonzero coefficient? Because then the other vectors in the equation all come *earlier* in the list. That is a stronger statement than "v_j is a combination of the others", and later proofs need it: when a list starts with vectors we want to keep, the redundant vector can be found after them.

Example: $(1,0), (0,1), (1,1), (2,0)$. One relation is $1 \cdot (1,0) + 1 \cdot (0,1) - 1 \cdot (1,1) + 0 \cdot (2,0) = 0$. The last nonzero coefficient belongs to $(1,1)$, and $(1,1) = (1,0) + (0,1)$ is in the span of the vectors before it.

Proof. 1. Take $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ with not all $\lambda_k = 0$. Let j be the largest index with $\lambda_j \neq 0$. The relation is $\lambda_1 v_1 + \cdots + \lambda_j v_j = 0$.

2. Divide by λ_j: $v_j = -\frac{\lambda_1}{\lambda_j} v_1 - \cdots - \frac{\lambda_{j-1}}{\lambda_j} v_{j-1} \in \operatorname{span}(v_1, \dots, v_{j-1})$. (If $j = 1$ this says $v_1 = 0$, and span of the empty list is $\{0\}$.)

3. For (ii): write $v_j = c_1 v_1 + \cdots + c_{j-1} v_{j-1}$. In any combination $\lambda_1 v_1 + \cdots + \lambda_m v_m$, replace v_j by this expression. The result uses only the other vectors. So $\operatorname{span}(v_1, \dots, v_m) \subseteq \operatorname{span}(\text{list without } v_j)$; the other inclusion is clear.

□

Proposition (Prop. 9 / LADR 2.30: a spanning list contains a basis)

If $f_1, \dots, f_n$ spans V, some of the f_k can be deleted so that what remains is a basis of V.

Proof. 1. If the list is independent, it is a basis.

2. If not, Prop. 8 gives a vector whose removal does not change the span. Remove it. The list still spans V and is one shorter.

3. Repeat. The list is finite, so this stops, and it stops only when the list is independent.

□

Theorem (Prop. 10 / LADR 2.22: independent lists are not longer than spanning lists)
If $e_1, \dots, e_m$ is linearly independent in V and $f_1, \dots, f_n$ spans V, then $m \le n$.

The idea

What has to be explained. Why can $\mathbb{R}^2$ not contain three independent vectors? There is no formula to compute; we need a mechanism that compares two lists.

The mechanism: one in, one out. Start with the spanning list $f_1, \dots, f_n$ and put e_1 into it. Since the f's span, e_1 is a combination of them, so the longer list $e_1, f_1, \dots, f_n$ is dependent, and one vector can be removed without changing the span. We want to remove an f, not e_1. That is possible: $e_1 = \lambda_1 f_1 + \cdots + \lambda_n f_n$ with some $\lambda_k \neq 0$ (because $e_1 \neq 0$), and that f_k can be solved for. Now we have a spanning list with e_1 and $n-1$ of the f's.

Repeat with e_2, e_3, ... Each time, the new e_k is a combination of the current list. Could that combination use only the e's already inserted? No: that would make $e_1, \dots, e_k$ dependent. So it uses some remaining f with a nonzero coefficient, and that f can be exchanged for e_k.

Why this gives $m \le n$. Each e that goes in uses up one f. If there were more e's than f's, the f's would run out: after n steps the list $e_1, \dots, e_n$ spans V, so e_{n+1} is a combination of $e_1, \dots, e_n$, contradicting independence.

Example in $\mathbb{R}^2$ with $e_1 = (1,1)$, $e_2 = (1,-1)$ and $f_1 = (1,0)$, $f_2 = (0,1)$: $e_1 = f_1 + f_2$, so exchange f_1: the list e_1, f_2 spans. Then $e_2 = e_1 - 2f_2$ uses f_2, so exchange it: e_1, e_2 spans. A third independent vector is impossible.

Proof. If $m = 0$ there is nothing to prove. Claim $P(k)$, for $1 \le k \le \min(m,n)$: after relabelling the f's, the list $L_k = (e_1, \dots, e_k, f_{k+1}, \dots, f_n)$ spans V.

1. $P(1)$: write $e_1 = \lambda_1 f_1 + \cdots + \lambda_n f_n$. Since $e_1 \neq 0$, some λ is nonzero; relabel so that $\lambda_1 \neq 0$. Then $f_1 = \frac{1}{\lambda_1}(e_1 - \lambda_2 f_2 - \cdots - \lambda_n f_n)$ is in $\operatorname{span}(e_1, f_2, \dots, f_n)$, so that span contains all the f's and is V.
2. $P(k-1) \Rightarrow P(k)$: since L_{k-1} spans, $e_k = \lambda_1 e_1 + \cdots + \lambda_{k-1} e_{k-1} + \mu_k f_k + \cdots + \mu_n f_n$.
3. Some μ is nonzero: otherwise $e_k \in \operatorname{span}(e_1, \dots, e_{k-1})$, contradicting independence of the e's. Relabel so that $\mu_k \neq 0$.
4. Solve for f_k in terms of $e_1, \dots, e_k, f_{k+1}, \dots, f_n$. So $\operatorname{span}(L_k)$ contains every vector of L_{k-1}, and L_k spans V.
5. Suppose $m > n$. Then $P(n)$ says $e_1, \dots, e_n$ spans V, so $e_{n+1} \in \operatorname{span}(e_1, \dots, e_n)$, contradicting independence. Hence $m \le n$.

The proof also shows: after all the e's are inserted, $e_1, \dots, e_m, f_{m+1}, \dots, f_n$ still spans V. □

Proposition (Prop. 11 / LADR 2.32: an independent list extends to a basis)
If V is finite-dimensional and $e_1, \dots, e_m$ is linearly independent, there is a basis of V containing $e_1, \dots, e_m$. If $f_1, \dots, f_n$ spans V, the added vectors can be taken from the f_j.

The idea

Put the two lists together: $e_1, \dots, e_m, f_1, \dots, f_n$ spans V. Now remove redundant vectors as in Prop. 9. The only worry is that an e gets removed. The dependence lemma removes a vector that is a combination of the vectors *before* it, and the e's come first and are independent, so no e is a combination of the vectors before it. Only f's are removed.

Proof. 1. The list $e_1, \dots, e_m, f_1, \dots, f_n$ spans V.

2. If it is dependent, Prop. 8 gives a vector in the span of the earlier ones. It is not some e_i, because $e_i \notin \operatorname{span}(e_1, \dots, e_{i-1})$ by independence. So it is an f_j; remove it. The span is unchanged.
3. Repeat until the list is independent. It still spans V and still contains every e_i.

□

- Ex. 11: no three vectors in $\mathbb{F}^2$ are independent, since e_1, e_2 spans. This is why three lines in $\mathbb{R}^2$ never form a direct sum.

- **Every subspace has a complement (LADR 2.33).** If U is a subspace of finite-dimensional V, there is a subspace W with $V = U \oplus W$: extend a basis $u_1, \dots, u_k$ of U to a basis $u_1, \dots, u_k, w_1, \dots, w_l$ of V and let $W = \operatorname{span}(w_1, \dots, w_l)$. Then $U + W = V$, and $U \cap W = \{0\}$: a vector in both is $\sum a_i u_i = \sum b_j w_j$, and independence of the whole basis forces all coefficients to be 0. (That U has a basis is Prop. 14 below.)

Infinite-dimensional spaces (Def. 9, optional in the notes). For an infinite set $M \subseteq V$: M is linearly independent if every finite subset is; M spans if every vector is a *finite* combination of elements of M; a basis is both.

- Ex. 12: $\{1, x, x^2, \dots\}$ is a basis of $\mathcal{P}(\mathbb{R})$.
- Ex. 13: in $\mathbb{F}^\infty$, $\{e_1, e_2, \dots\}$ is independent but not a basis: $(1, 1, 1, \dots)$ is not a finite combination (HW2 P2).
- Every vector space has a basis, but the proof for infinite-dimensional spaces needs Zorn's lemma.

§2.2 Dimension (Ch. 4)

Proposition (Prop. 12 / LADR 2.34, Def. 10)

Any two bases of a finite-dimensional V have the same length. This length is the **dimension** $\dim V$.

Proof. Let B_1, B_2 be bases. B_1 is independent and B_2 spans, so length $B_1 \le$ length B_2 (Prop. 10). Swap the roles for the other inequality. □

- Ex. 14–16: $\dim \mathbb{F}^n = n$. $\dim \mathcal{P}_r(\mathbb{F}) = r + 1$ (basis $1, x, \dots, x^r$). $\dim\{0\} = 0$.
- Ex. 17: the dimension depends on the field. $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$, but $\dim_{\mathbb{R}} \mathbb{C}^2 = 4$, with real basis $(1, 0), (i, 0), (0, 1), (0, i)$.

Proposition (Prop. 13 / LADR 2.38, 2.42: the right number of vectors)

Let $d = \dim V$. (i) Any linearly independent list of d vectors is a basis. (ii) Any spanning list of d vectors is a basis.

The idea

This is the counting principle from Week 1 (a map between finite sets of equal size is injective exactly when it is surjective), with dimension in place of the number of elements. An independent list of d vectors has no room to grow; a spanning list of d vectors has nothing to lose. On an exam it halves the work: with the right number of vectors, check only one of the two properties.

Proof. 1. Extend the independent list to a basis (Prop. 11). Every basis has length d, so nothing was added.
2. Reduce the spanning list to a basis (Prop. 9). Every basis has length d, so nothing was removed.

□

Proposition (Prop. 14 / LADR 2.25, 2.37, 2.39: subspaces)

If U is a subspace of finite-dimensional V, then U is finite-dimensional, $\dim U \le \dim V$, and $\dim U = \dim V \iff U = V$.

Proof. 1. Independent lists in U are independent in V, so they have length $\le \dim V$ (Prop. 10). Choose an independent list $u_1, \dots, u_k$ in U of the largest possible length.
2. It spans U: if some $u \in U$ were not in its span, $u_1, \dots, u_k, u$ would be independent (in a relation the coefficient of u must be 0, and then the others are), contradicting maximality. So it is a basis of U, and $\dim U = k \le \dim V$.
3. If $\dim U = \dim V$, this basis of U is an independent list of $\dim V$ vectors in V, so it is a basis of V (Prop. 13). Then $U = \operatorname{span} = V$.

□

- Ex. 18: $1, x-a, (x-a)^2, \ldots, (x-a)^n$ is a basis of $\mathcal{P}_n(\mathbb{R})$. Independent: the degrees are different, so in a relation the highest-degree term with nonzero coefficient cannot cancel. And there are $n+1 = \dim \mathcal{P}_n(\mathbb{R})$ of them (Prop. 13). The coordinates are the Taylor coefficients: $p(x) = \sum_k \frac{p^{(k)}(a)}{k!}(x-a)^k$.
- Finite dimension is needed: $\mathcal{P}(\mathbb{R})$ has the proper subspace $\{p : p(0) = 0\}$, and both are infinite-dimensional.

Theorem (Prop. 15, 16 / LADR 2.43: dimension of a sum)

For subspaces U, W of a finite-dimensional space:

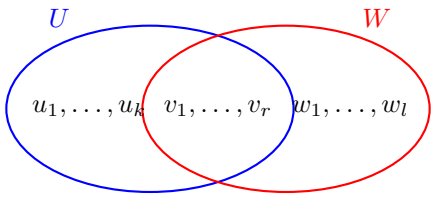
$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W).$$

More precisely, if $v_1, \ldots, v_r$ is a basis of $U \cap W$, it extends to a basis $v_1, \ldots, v_r, u_1, \ldots, u_k$ of U and to a basis $v_1, \ldots, v_r, w_1, \ldots, w_l$ of W, and then $v_1, \ldots, v_r, u_1, \ldots, u_k, w_1, \ldots, w_l$ is a basis of $U+W$.

The idea

For finite sets, $|A \cup B| = |A| + |B| - |A \cap B|$: adding the two sizes counts the common part twice. The same thing happens with bases, provided the bases of U and W are chosen so that their common part is a basis of $U \cap W$. That is why the proof *starts* from a basis of $U \cap W$ and extends it in two directions.

Putting the three groups together clearly spans $U+W$. The step that needs an argument is independence, and the question to ask is: in a relation, could the u-part and the w-part cancel each other? Move the w-part to the other side. Then one vector is written both as an element of U and as an element of W, so it lies in $U \cap W$ and is a combination of the v's alone. That forces the u-coefficients to be 0, and then the rest.



$$\dim(U+W) = r + k + l = (r+k) + (r+l) - r$$

A basis of $U \cap W$ (middle) extended to a basis of U (left two groups) and of W (right two groups). All three groups together are a basis of $U+W$.

Proof. 1. Extend $v_1, \ldots, v_r$ to bases of U and of W as stated (Prop. 11 in U and in W).
2. The combined list spans $U+W$: every $u+w$ is a combination of it.
3. Independence. Suppose $\sum \lambda_i v_i + \sum \mu_j u_j + \sum \nu_t w_t = 0$. Then $x = \sum \lambda_i v_i + \sum \mu_j u_j = -\sum \nu_t w_t$. The first expression is in U, the second in W, so $x \in U \cap W$.
4. So $x = \sum \alpha_i v_i$ for some α_i. Then $\sum \alpha_i v_i + \sum \nu_t w_t = 0$ is a relation in the basis of W, so every $\nu_t = 0$.
5. The original relation is now $\sum \lambda_i v_i + \sum \mu_j u_j = 0$, a relation in the basis of U, so every $\lambda_i = 0$ and $\mu_j = 0$.
6. Count: $\dim(U \cap W) = r$, $\dim U = r+k$, $\dim W = r+l$, $\dim(U+W) = r+k+l$.

□

Proposition (Complement criterion)

For subspaces U, W of a finite-dimensional V, any two of the following imply the third (and then $V = U \oplus W$):

(a) $\dim U + \dim W = \dim V$ (b) $U \cap W = \{0\}$ (c) $U + W = V$.

Proof. By the theorem, $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$. (a),(b) give $\dim(U+W) = \dim V$, so $U+W = V$ (Prop. 14). (a),(c) give $\dim(U \cap W) = 0$. (b),(c) give $\dim V = \dim U + \dim W$. □

Example (Ex. 19: two planes in $\mathbb{F}^3$)

$U = \operatorname{span}\big((1,1,0),(0,1,1)\big)$, $W = \operatorname{span}\big((1,2,2),(2,2,1)\big)$.

1. Is $(1,2,2) \in U$? $(1,2,2) = \lambda(1,1,0) + \mu(0,1,1)$ needs $\lambda = 1$ (first coordinate) and $\mu = 2$ (third), but then the second coordinate is $3 \neq 2$. No.
2. So $(1,1,0),(0,1,1),(1,2,2)$ are independent: three independent vectors in $\mathbb{F}^3$, a basis (Prop. 13). Hence $U + W = \mathbb{F}^3$.
3. $\dim(U \cap W) = 2 + 2 - 3 = 1$: the planes meet in a line.
4. To find the line, solve $a(1,1,0)+b(0,1,1) = c(1,2,2)+d(2,2,1)$: $a = c+2d$, $a+b = 2c+2d$, $b = 2c+d$. Adding the first and third: $a+b = 3c+3d$, so $c+d = 0$. Take $c = 1$, $d = -1$: the vector is $(1,2,2)-(2,2,1) = (-1,0,1)$. So $U \cap W = \operatorname{span}\big((-1,0,1)\big)$.

- Ex. 20: for three subspaces apply the formula twice: $\dim(U_1+U_2+U_3) = \dim U_1 + \dim U_2 + \dim U_3 - \dim(U_1 \cap U_2) - \dim((U_1+U_2)\cap U_3)$. There is no formula using only the pairwise intersections: three lines in $\mathbb{F}^2$ have all pairwise intersections $\{0\}$, and $\dim(U_1+U_2+U_3) = 2 \neq 3$.

Proposition (Prop. 17 / LADR 3.94)

If the sum $U_1 \oplus \cdots \oplus U_m$ is direct, then $\dim(U_1 \oplus \cdots \oplus U_m) = \dim U_1 + \cdots + \dim U_m$.

Proof. Induction on m. By the theorem, $\dim((U_1+\cdots+U_{m-1})+U_m) = \dim(U_1+\cdots+U_{m-1}) + \dim U_m - \dim\big((U_1+\cdots+U_{m-1})\cap U_m\big)$, and the last intersection is $\{0\}$ by the definition of a direct sum (W2). □

Computing with lists in $\mathbb{F}^n$.

1. *Independent?* Put the vectors as the columns of a matrix and row reduce. Independent $\iff$ every column has a pivot. The columns with pivots always form a basis of the span.
2. *Basis of $U+W$:* put the two bases together and delete redundant vectors (Prop. 9).
3. *Basis of $U \cap W$:* set a general combination of the basis of U equal to a general combination of the basis of W and solve, as in Ex. 19. First get its dimension from the formula so you know how many vectors to expect.

Shortcuts

- **Right number of vectors**: with $\dim V$ vectors, check only independence or only spanning (Prop. 13).
- **Too many or too few**: more than $\dim V$ vectors are dependent; fewer than $\dim V$ do not span (Prop. 10). No computation.
- **Two vectors** are dependent exactly when one is a multiple of the other.
- **Polynomials of different degrees** are independent.
- **Direct sum by counting**: if $U + W = V$ and $\dim U + \dim W = \dim V$, the sum is direct (complement criterion); no need to compute $U \cap W$ (HW2 P5).
- **Equality of subspaces**: $U \subseteq W$ and $\dim U = \dim W$ give $U = W$ (Prop. 14).
- **Subspace given by k independent linear conditions** in an n-dimensional space: expect dimension $n-k$ (proved next week by rank–nullity); use this to know when you have found enough basis vectors.

Traps

- Independence of each list separately says nothing about sums: v and $-v$ are each independent lists, but $v + (-v) = 0$ (HW2 P1).
- "v_j is not a multiple of any other vector in the list" does not give independence for three or more vectors: $(1,0), (0,1), (1,1)$.
- A basis of V need not contain a basis of a given subspace U: e_1, e_2 contains no basis of $\operatorname{span}((1,1))$.
- $\dim(U_1 + U_2 + U_3)$ has no inclusion–exclusion formula.
- The dimension depends on the field: $\dim_{\mathbb{R}} \mathbb{C} = 2$.
- In infinite dimensions an independent set need not extend to a basis by adding nothing: $\{e_1, e_2, \dots\}$ in $\mathbb{F}^\infty$ does not span.

§3 Problem set

HW2 P1 (LADR 2A.11). Prove or give a counterexample: if $v_1, \dots, v_n$ and $w_1, \dots, w_n$ are linearly independent lists in V, then $v_1 + w_1, \dots, v_n + w_n$ is linearly independent.

Hint: try $n = 1$.

HW2 P2. Let $\mathbb{F}^\infty$ be the space of all sequences $(a_1, a_2, \dots)$, $a_j \in \mathbb{F}$, and e_j the sequence with 1 in position j and 0 elsewhere. (a) Verify that $\mathbb{F}^\infty$ is a vector space. (b) Prove that it is infinite-dimensional. (c) Prove that $e_1, e_2, \dots$ do not span $\mathbb{F}^\infty$.

Hint: (b) Prop. 10: a finite spanning list bounds the length of independent lists. (c) a finite combination has only finitely many nonzero entries.

HW2 P3 (LADR 2B.5). Suppose V is finite-dimensional and U, W are subspaces with $V = U + W$. Prove that V has a basis all of whose vectors belong to $U \cup W$.

Hint: a spanning list contains a basis (Prop. 9).

HW2 P4. (a) Prove that every nonzero proper subspace of $\mathbb{F}^2$ has the form $\operatorname{span}(v)$ for some $v \neq 0$. (b) Let $v, w \in \mathbb{F}^2$. Prove that v, w is a basis of $\mathbb{F}^2$ if and only if neither is a scalar multiple of the other.

Hint: dimensions of subspaces (Prop. 14) and Prop. 13.

HW2 P5 (LADR 2C.12). Suppose U, W are subspaces of $\mathbb{R}^8$ with $\dim U = 3$, $\dim W = 5$ and $U + W = \mathbb{R}^8$. Prove that $\mathbb{R}^8 = U \oplus W$.

Hint: the dimension formula.

HW2 P6. Let $U = \{p \in \mathcal{P}_4(\mathbb{R}) : p(1) = p'(1) = 0\}$. (a) Prove that U is a subspace. (b) Find a basis of U and determine $\dim U$ directly, without rank–nullity.

Hint: write p in the basis $1, (x-1), \dots, (x-1)^4$ of Ex. 18.

R1 (Practice Midterm 1, Question 1). Let $u_1 = (1,1,0,0)$, $u_2 = (0,1,1,0)$, $w_1 = (0,0,1,1)$, $w_2 = (1,0,0,-1)$, $U = \operatorname{span}(u_1, u_2)$, $W = \operatorname{span}(w_1, w_2)$. (a) Prove that $\mathbb{R}^4 = U \oplus W$. (b) Write $(2,1,3,0)$ as $u + w$ with $u \in U$, $w \in W$.

R2 (Midterm 1 review, Exercise 1). Let U be a subspace of a finite-dimensional V. Prove that there is a subspace W with $U \oplus W = V$.

R3 (Midterm 1 review, Exercise 3). Let $V = \{p \in \mathcal{P}_2(\mathbb{R}) : p'(1) = p(2)\}$. Construct a basis of V and prove it is a basis.